

Qualification Pack



Assistant Technician-Drilling (Oil & Gas)

QP Code: HYC/Q0101

Version: 4.0

NSQF Level: 4

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HYC/Q0101: Assistant Technician-Drilling (Oil & Gas)

Brief Job Description

The Assistant Technician-Drilling (Oil & Gas) is to carry out the drilling operation task like Rig building, Rig dismantling, Top man operation, Rig floor, operation, cementation job and other drilling activities and operate Drill floor operations to drill shallow as well as deep holes for exploring and exploiting Oil & Gas. This operation is a specialized task that can be safely perform only with adequate training and experience. The individual should have knowledge for performing basic maintenance of the machine.

Personal Attributes

This job requires the individual to have good problem-solving Skills, knowledge of the Hydrocarbon Sector, Drilling process and systems would be an added advantage. Additional skills, including communication skills strong customer service, strong problem-solving skills should be self-motivated should be able to apply practical judgement to successfully perform the assigned responsibilities and a team player with ability to work with team and under pressure

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N0101: Perform Drilling Operation](#)
2. [HYC/N0102: Occupational health and safety \(OHAS\)](#)
3. [HYC/N0103: Work effectively with colleagues and supervisor](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Upstream
Occupation	Exploration and Production
Country	India
NSQF Level	4
Credits	20

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/8113.0100
Minimum Educational Qualification & Experience	10th Class (with minimum 2 years of relevant experience) OR 12th Class (Science) OR I.T.I (in engineering trade after class 10th) OR Diploma (3-years Diploma in relevant field)
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	Knowledge of basic machine tools handling would be an added advantage
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/05/2028
NSQC Approval Date	08/05/2025
Version	4.0
Reference code on NQR	QG-04-HY-04020-2025-V2-HSSCI
NQR Version	4.0

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HYC/N0101: Perform Drilling Operation

Description

This unit is about performing drilling operations at Oil & Gas well site.

Scope

The scope covers the following :

- This unit is about performing drilling operations at Oil & Gas well site.

Elements and Performance Criteria

Drilling Process

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding and knowledge of the types of Oil & Gas Wells Exploratory well Developmental well Geopressure/geothermal well Stripper well Multiple completion well Injection well Service well
- PC2.** Understand Platform types for underwater drilling submersible barges and platforms Shallow water complex Gravity base Compliant towers FPSO: Floating Production, Storage and Offloading Floating platforms Jack-ups (on legs) Tension Leg Platform (TLP) Semi-submersible platforms Drilling barges Drill ships Fixed on site platforms Sub-sea templates
- PC3.** Understand the on shore well components
- PC4.** Identify Upstream process sections: Wellhead, Manifolds and gathering, Separation, Metering, storage and export and Utility systems
- PC5.** Understanding the Components of Drilling Rig (derrick, floor, drawworks, drive and mud handling, drill string, Drill Collar, Blowout Preventer, turn table, Crown Block, Top drive, Iron Roughneck, Mud Pumps.)
- PC6.** Understanding the following components of well i.e. Well casing - Conductor casing, Surface casing, Intermediate casing, Production casing. Use of tubing and Packer and Completion (Open hole completions, Conventional perforated completions, Sand exclusion completions, Permanent completions, Multiple zone completion, Drainhole completions)
- PC7.** Understanding of working at height, operation at monkey board, top man escape device
- PC8.** Knowledge of the Wellhead Dry or subsea completion Casing head and casing hangers The tubing hanger Master gate valve Pressure gauge Wing valve Swab valve Variable flow choke valve Vertical tree Subsea wells (umbilical) Injection
- PC9.** Perform Artificial lift operation: Rod pumps, electrical submerged pump (ESP), Gas lift and Plunger lift
- PC10.** Knowledge of the Well workover, intervention and stimulation
- PC11.** Understanding Manifolds and gathering Pipelines and risers Production, test and injection manifolds

Drilling Techniques

To be competent, the user/individual on the job must be able to:

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- PC12.** Understanding/knowledge of the Drilling Methods Drilling rigs Percussion or cable drilling Rotary drilling Rotary percussion drilling Electro and turbo drilling Directional drilling explosive drilling and flame piercing
- PC13.** Perform Abandonment Process
- PC14.** Perform drilling Operations Drilling techniques Drilling mud Casing and cementation
- PC15.** Understanding Completion, Enhanced Recovery and Workover Operations
- PC16.** Understand a Typical Drilling Program

Basic Maintenance Techniques and Practices

To be competent, the user/individual on the job must be able to:

- PC17.** Prepare for basic routine maintenance tasks
- PC18.** Describe basic routine maintenance techniques
- PC19.** Carry out basic routine maintenance tasks
- PC20.** Carry out basic inspections
- PC21.** Identify the use of basic maintenance hand tools
- PC22.** Demonstrate the use of basic hand tools for maintenance tasks and the production of an artifact
- PC23.** Demonstrate the care of basic maintenance hand tools

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisational sop on drilling process
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** escalation matrix for reporting identified incidents, troubles and/ or emergencies e.g. system failures ,fire and power failures
- KU4.** records to be maintained and implications of non-maintenance of the same
- KU5.** she and ohs guidelines and regulations as per companys norms
- KU6.** basic oil & gas drilling process and operations
- KU7.** basic maintenance hand tools
- KU8.** basic measuring and marking-out instruments
- KU9.** basic engineering communications
- KU10.** basic numerical skills
- KU11.** physical science and basic properties of materials

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up appropriate technical forms, activity logs in required format of the company
- GS2.** maintain proper records as per given format
- GS3.** read and understand manuals, work orders, health and safety instructions, memos, reports etc.

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- GS4.** cautious to the customers
- GS5.** liaisoning and coordination skills
- GS6.** communicate with supervisor and peers
- GS7.** communicate in the local language (preferable)
- GS8.** plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS9.** think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS10.** identify immediate or temporary solutions to resolve delays
- GS11.** identify sources of support that can be availed of for problem solving for various kind of problems
- GS12.** seek appropriate assistance from other sources to resolve problems
- GS13.** report problems that you cannot resolve to appropriate authority
- GS14.** identify defects in the material and communicate it at the earliest and suggest
- GS15.** improvements(if any) in process/material based on experience
- GS16.** apply problem-solving approaches in different situations
- GS17.** refer anomalies to the line manager sb18. identify any issues affecting the material, equipment or surroundings
- GS18.** escalate issues that cannot be solved as per the troubleshooting/company manual

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Drilling Process</i>	9	38	-	-
PC1. Understanding and knowledge of the types of Oil & Gas Wells Exploratory well Developmental well Geopressure/geothermal well Stripper well Multiple completion well Injection well Service well	-	5	-	-
PC2. Understand Platform types for underwater drilling submersible barges and platforms Shallow water complex Gravity base Compliant towers FPSO: Floating Production, Storage and Offloading Floating platforms Jack-ups (on legs) Tension Leg Platform (TLP) Semi-submersible platforms Drilling barges Drill ships Fixed on site platforms Sub-sea templates	-	5	-	-
PC3. Understand the on shore well components	-	5	-	-
PC4. Identify Upstream process sections: Wellhead, Manifolds and gathering, Separation, Metering, storage and export and Utility systems	1	4	-	-
PC5. Understanding the Components of Drilling Rig (derrick, floor, drawworks, drive and mud handling, drill string, Drill Collar, Blowout Preventer, turn table, Crown Block, Top drive, Iron Roughneck, Mud Pumps.)	1	4	-	-
PC6. Understanding the following components of well i.e. Well casing - Conductor casing, Surface casing, Intermediate casing, Production casing. Use of tubing and Packer and Completion (Open hole completions, Conventional perforated completions, Sand exclusion completions, Permanent completions, Multiple zone completion, Drainhole completions)	1	4	-	-
PC7. Understanding of working at height, operation at monkey board, top man escape device	2	4	-	-
PC8. Knowledge of the Wellhead Dry or subsea completion Casing head and casing hangers The tubing hanger Master gate valve Pressure gauge Wing valve Swab valve Variable flow choke valve Vertical tree Subsea wells (umbilical) Injection	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Perform Artificial lift operation: Rod pumps, electrical submerged pump (ESP), Gas lift and Plunger lift	1	1	-	-
PC10. Knowledge of the Well workover, intervention and stimulation	1	1	-	-
PC11. Understanding Manifolds and gathering Pipelines and risers Production, test and injection manifolds	1	1	-	-
<i>Drilling Techniques</i>	7	14	-	-
PC12. Understanding/knowledge of the Drilling Methods Drilling rigs Percussion or cable drilling Rotary drilling Rotary percussion drilling Electro and turbo drilling Directional drilling explosive drilling and flame piercing	1	4	-	-
PC13. Perform Abandonment Process	1	2	-	-
PC14. Perform drilling Operations Drilling techniques Drilling mud Casing and cementation	1	4	-	-
PC15. Understanding Completion, Enhanced Recovery and Workover Operations	1	2	-	-
PC16. Understand a Typical Drilling Program	3	2	-	-
<i>Basic Maintenance Techniques and Practices</i>	18	14	-	-
PC17. Prepare for basic routine maintenance tasks	3	2	-	-
PC18. Describe basic routine maintenance techniques	2	2	-	-
PC19. Carry out basic routine maintenance tasks	4	1	-	-
PC20. Carry out basic inspections	1	2	-	-
PC21. Identify the use of basic maintenance hand tools	3	2	-	-
PC22. Demonstrate the use of basic hand tools for maintenance tasks and the production of an artifact	3	2	-	-
PC23. Demonstrate the care of basic maintenance hand tools	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	34	66	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N0101
NOS Name	Perform Drilling Operation
Sector	Hydrocarbon
Sub-Sector	Upstream
Occupation	Exploration and Production
NSQF Level	4
Credits	13
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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HYC/N0102: Occupational health and safety (OHAS)

Description

This unit is about Health and Safety in Production and Maintenance

Scope

The scope covers the following :

- This unit is about Health and Safety in Production and Maintenance

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks like cutting, welding, repairing in pipeline jobs, well operations, maintenance inside the installation and work conditions during day to day work and during emergency.
- PC2.** state the name and location of people responsible for health and safety for the workplace and escalation matrix.
- PC3.** identify job-site hazardous work and state possible causes of risk or possible accidents in the workplace.
- PC4.** carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role.
- PC5.** state location of general health and safety equipment in the workplace.
- PC6.** inspect for faults set up and safely use steps and ladders in general use
- PC7.** work safely in and around trenches, elevated places and confined areas
- PC8.** identify common risks and safety sop in oil & gas production area

Fire safety

To be competent, the user/individual on the job must be able to:

- PC9.** use the various appropriate fire extinguishers on different types of fires correctly.
- PC10.** identify and follow pro active and reactive fire fighting sop in oil & gas production facilities
- PC11.** perform fire evacuation steps
- PC12.** prepare incident reports

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC13.** availability of first aid box & accessories
- PC14.** demonstrate how to free a person from electrocution
- PC15.** administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC16.** administer appropriate first aid in chemical hazard
- PC17.** perform and organize loss minimization or rescue activity during an accident in real or simulated environments

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- PC18.** follow sop in oil & gas production facilities
- PC19.** use/propoer utilization of breathing apparatus
- PC20.** ensure emergency preparation and resonse

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
- KU2.** names and location of documents that refer to health and safety in the workplace.
- KU3.** risk and impact of not following defined procedures/work instructions
- KU4.** escalation matrix for reporting identified incidents, troubles and/ or emergencies e.g. system failures ,fire and power failures
- KU5.** records to be maintained and implications of non-maintenance of the same
- KU6.** she and ohs guidelines and regulations as per companys norms
- KU7.** understanding of 5s and housekeeping
- KU8.** occupational health and safety
- KU9.** hazards and its effects and risk assessment
- KU10.** safe working practices
- KU11.** use of different types of personalprotectiveequipments(ppe)
- KU12.** various types of fire fighting equipments& its uses in different conditions
- KU13.** standard operating procedures in oil and gas safety w.r.t. occupational health & safety
- KU14.** risk management and reporting

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up appropriate technical forms, activity logs in required format of the company
- GS2.** maintain proper records as per given format
- GS3.** read and understand manuals, work orders, health and safety instructions, memos, reports etc.
- GS4.** cautious to the customers
- GS5.** liaisioning and coordination skills
- GS6.** communicate with supervisor and peers
- GS7.** communicate in the local language (preferable)
- GS8.** plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS9.** think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS10.** identify immediate or temporary solutions to resolve delays

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- GS11.** identify sources of support that can be availed of for problem solving for various kind of problems
- GS12.** seek appropriate assistance from other sources to resolve problems
- GS13.** report problems that you cannot resolve to appropriate authority
- GS14.** identify defects in the material and communicate it at the earliest and suggest
- GS15.** apply problem-solving approaches in different situations
- GS16.** refer anomalies to the line manager
- GS17.** escalate issues that cannot be solved as per the troubleshooting/company manual

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	16	27	-	-
PC1. use protective clothing/equipment for specific tasks like cutting, welding, repairing in pipeline jobs, well operations, maintenance inside the installation and work conditions during day to day work and during emergency.	3	3	-	-
PC2. state the name and location of people responsible for health and safety for the workplace and escalation matrix.	-	5	-	-
PC3. identify job-site hazardous work and state possible causes of risk or possible accidents in the workplace.	3	3	-	-
PC4. carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role.	3	3	-	-
PC5. state location of general health and safety equipment in the workplace.	-	5	-	-
PC6. inspect for faults set up and safely use steps and ladders in general use	3	2	-	-
PC7. work safely in and around trenches, elevated places and confined areas	3	2	-	-
PC8. identify common risks and safety sop in oil & gas production area	1	4	-	-
<i>Fire safety</i>	11	10	-	-
PC9. use the various appropriate fire extinguishers on different types of fires correctly.	2	3	-	-
PC10. identify and follow pro active and reactive fire fighting sop in oil & gas production facilities	3	2	-	-
PC11. perform fire evacuation steps	3	3	-	-
PC12. prepare incident reports	3	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Emergencies, rescue and first-aid procedures</i>	15	21	-	-
PC13. availability of first aid box & accessories	2	3	-	-
PC14. demonstrate how to free a person from electrocution	3	2	-	-
PC15. administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.	2	3	-	-
PC16. administer appropriate first aid in chemical hazard	2	3	-	-
PC17. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	2	3	-	-
PC18. follow sop in oil & gas production facilities	2	3	-	-
PC19. use/propoer utilization of breathing apparatus	1	2	-	-
PC20. ensure emergency preparation and resonse	1	2	-	-
NOS Total	42	58	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N0102
NOS Name	Occupational health and safety (OHAS)
Sector	Hydrocarbon
Sub-Sector	Upstream
Occupation	Exploration and Production
NSQF Level	4
Credits	3
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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HYC/N0103: Work effectively with colleagues and supervisor

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace.

Scope

The scope covers the following :

- This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace.

Elements and Performance Criteria

Work effectively with colleagues and supervisor

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** demonstrate time management skills
- PC11.** understands expectation management
- PC12.** demonstrate commitment to work, adhering to sops, honesty etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** risk and impact of not following defined procedures/work instructions

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- KU4.** escalation matrix for reporting identified incidents, troubles and/ or emergencies e.g. system failures ,fire and power failures
- KU5.** records to be maintained and implications of non-maintenance of the same
- KU6.** she and ohs guidelines and regulations as per companys norms
- KU7.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU8.** importance of effective communication in the workplace
- KU9.** importance of teamwork in organizational and individual success
- KU10.** various components of effective communication
- KU11.** key elements of active listening
- KU12.** value and importance of active listening and assertive communication
- KU13.** barriers to effective communication
- KU14.** importance of tone and pitch in effective communication
- KU15.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU16.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU17.** importance of ethics for professional success
- KU18.** importance of discipline for professional success
- KU19.** constitutes disciplined behavior for a working professional

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up appropriate technical forms, activity logs in required format of the company
- GS2.** maintain proper records as per given format
- GS3.** read and understand manuals, work orders, health and safety instructions, memos, reports etc.
- GS4.** corteous to the peers , juniors , superiors and customers
- GS5.** liaisoning and coordination skills
- GS6.** make decisions on a suitable course of action or response
- GS7.** plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS8.** think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS9.** identify immediate or temporary solutions to resolve delays
- GS10.** identify sources of support that can be availed of for problem solving for various kind of problems
- GS11.** seek appropriate assistance from other sources to resolve problems
- GS12.** report problems that you cannot resolve to appropriate authority
- GS13.** identify defects in the material and communicate it at the earliest and suggest

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- GS14.** apply problem-solving approaches in different situations
- GS15.** refer anomalies to the line manager sb18. identify any issues affecting the material, equipment or surroundings
- GS16.** escalate issues that cannot be solved as per the troubleshooting/company manual

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with colleagues and supervisor</i>	22	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	2	6	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	2	6	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	2	6	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	-	4	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	2	6	-	-
PC6. display appropriate communication etiquette while working	2	6	-	-
PC7. display active listening skills while interacting with others at work	2	6	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	2	6	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	2	6	-	-
PC10. demonstrate time management skills	2	6	-	-
PC11. understands expectation management	2	6	-	-
PC12. demonstrate commitment to work, adhering to sops, honesty etc.	2	6	-	-
NOS Total	22	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N0103
NOS Name	Work effectively with colleagues and supervisor
Sector	Hydrocarbon
Sub-Sector	Upstream
Occupation	Exploration and Production
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N0101.Perform Drilling Operation	34	66	-	-	100	50
HYC/N0102.Occupational health and safety (OHAS)	42	58	-	-	100	20
HYC/N0103.Work effectively with colleagues and supervisor	22	70	-	-	92	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	118	224	-	-	342	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.